

CS471 – Web Technologies

Lab 1: The Internet Protocols

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Task 1

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	• Sets up a connection first (three-way handshake: SYN, SYN-ACK, ACK) • Confirms received data with ACK • Resends lost packets (retransmission) • UDP has no connection and no delivery check
Data Integrity and Ordering	TCP	• Uses sequence numbers to put data in order • Detects missing data and resends it • UDP: packets can be lost or arrive out of order

Task 2

	TCP	UDP
Use cases	• Web browsing (HTTP/HTTPS) • Email • File transfer (FTP) • SSH	• Live streaming • Voice/video calls (VoIP) • Online games • DNS
Performance	• Slower (handshake, ACKs, resending) • Reliable, data arrives correctly	• Faster (small header, no handshake) • Some packets may be lost